

Beyond Accuracy: A Clinical Validation Approach to Cardiovascular Disease Prediction Using Machine Learning

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DSE6311 Data Science Capstone

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May 2026

ABSTRACT:

This study applies a clinical validation framework to cardiovascular disease (CVD) prediction using machine learning, reframing model evaluation as a clinical risk question rather than a purely statistical one. Using the UCI Heart Disease dataset (n=297), four supervised classification models (Logistic Regression, Random Forest, XGBoost and Support Vector Machine) were trained, tuned, and evaluated using sensitivity, specificity, and Area Under the Receiver Operating Curve (AUC). Cost-Sensitive learning (CSL) was applied across all models to embed clinical risk directly into training by penalizing false negatives more heavily than false positives. A three-layer clinical validation framework was applied: standard metrics, threshold analysis, and Decision Curve Analysis (DCA). Contrary to the initial prediction, Logistic Regression with cost-sensitive learning outperformed all ensemble methods, consistent with published literature on small clinical datasets. At the default classification threshold of 0.50, no model achieved the clinical sensitivity target of $\geq 85\%$. Threshold optimization at 0.25 revealed a sensitivity of 94.9% while maintaining specificity above the clinical minimum of 75%. DCA confirmed clinical utility across all four tuned models, demonstrating that model guided decisions would produce better patient outcomes than treating all or treating no patients. These findings demonstrate that clinically appropriate model evaluation requires sensitivity, specificity, threshold analysis, and decision curve analysis evaluated together. Accuracy alone is insufficient for clinical support validation.

INTRODUCTION

Research Question:

This project evaluates machine learning model performance through a clinical validation lens rather than standard accuracy metrics. Given the trade-off between sensitivity and specificity, how should model performance be evaluated to assess clinically appropriate risk in the context of cardiovascular disease?

Background:

Cardiovascular disease (CVD) remains the leading cause of death for men and women globally, accounting for 19.8 million deaths annually (WHO, 2022). Gaps in detection persist due to socioeconomic constraints, underestimation of cardiovascular risk, and underuse of digital health tools such as AI-based machine learning (Dores et al., 2026). The published research on machine-learning-based CVD prediction is well represented in the literature, yet most published work stops at accuracy and AUC metrics. A model that achieves 92% accuracy but misses 40% of true positives would be statistically impressive but clinically dangerous. A recent review of machine learning studies on heart disease prediction supports this, finding that model selection should be based on performance metrics that are crucial for clinical diagnostic accuracy rather than overall accuracy alone (Kumar et al., 2025).

Platforms that detect coronary artery disease lesions using AI-driven diagnostic tools are becoming standard in clinical imaging environments. The ability to thoroughly assess model performance against clinical risk thresholds is a critical skill for clinical AI validation teams. Validation teams must not only know if a model is accurate, but whether the model's error profile can be clinically trusted to make it both academically rigorous and professionally meaningful.

The primary stakeholder identified is clinical AI and machine learning developers working in medical imaging and diagnostic AI. They are the teams responsible for assessing whether a model is safe for deployment in a clinical workflow. Secondary stakeholders include cardiologists and clinical informaticists who use and interpret these tools at the point of care.

Cardiovascular disease prediction using the UCI Heart Disease dataset is well used in machine learning publications. This project is not novel in that sense. The project will aim to reframe and improve the approach from a clinical validation standpoint and apply it to model evaluation.

Academic projects treat model evaluation as a report of sensitivity and specificity metrics.

Clinical AI development treats those same metrics as design decisions with real consequences for patient safety. This project applies that clinical standard to an academic context.

Hypothesis and Prediction:

Hypothesis: Optimizing a machine learning model's sensitivity will reduce false negatives but increase false positives. In cardiovascular disease prediction, this trade-off carries measurable clinical complications. A model cannot be evaluated as clinically appropriate without also analyzing what its error profile means in a real-world care setting. It is also important to acknowledge that false positives are not a trivial outcome. Overcalling healthy patients increases patient anxiety, drives up healthcare costs, and erodes patient trust in healthcare. The goal is not to minimize false negatives by any means necessary, but to find the threshold where the tradeoff is clinically defensible.

Prediction: A tuned ensemble model (Random Forest or XGBoost) optimized for sensitivity will outperform a baseline Logistic Regression model on recall and ROC-AUC. However, the most appropriate model will not necessarily be the highest performing model based on accuracy — it

will be the model whose sensitivity and specificity balance best aligns with defined clinical risk thresholds. Cost-sensitive learning will also be explored as a potential approach, assigning a higher misclassification penalty to false negatives during model training to reflect the reality that missing a true positive carries greater risk than a false positive (Brownlee, 2020). Sensitivity and specificity threshold goals are $\geq 85\text{-}90\%$ and $\geq 75\text{-}80\%$, respectively, grounded in benchmarks from comparable CVD classification studies (Vyshnya et al., 2024). Foundational clinical methods literature establishes that sensitivity and specificity must always be interpreted together with disease prevalence and clinical context rather than in isolation (Simon and Boring, 1990). It is important to note that this prediction was not supported by the final results. Logistic Regression with cost-sensitive learning outperformed all ensemble methods, however, the core elements were validated, as discussed in the findings and conclusions sections.

DATA ACQUISITION

Data Source

The UCI Heart Disease dataset is one of the most widely used datasets in cardiovascular machine learning research. The dataset was originally collected across four institutions. The dataset contains 76 attributes, although all published experiments have referred to a subset of 14 of them. The Cleveland database is the only one that has been used by machine learning researchers to date, and the experiments have focused on distinguishing between the presence and absence of disease. The dataset is formally cited as: Janosi, A., Steinbrunn, W., Pfisterer, M., & Detrano, R. (1989). Heart Disease [Dataset]. UCI Machine Learning Repository.

<https://doi.org/10.24432/C52P4X>.

For this project, the dataset was initially loaded from a public GitHub repository. Upon examining the unique values of categorical variables, two anomalies were detected. The variable “ca”, according to the original UCI documentation, should contain values 0-3, representing the number of major vessels identified by fluoroscopy, but it also contained a value of 4, which is not documented. The variable “thal”, which represents thalassemia type, should contain values 3, 6, and 7. This variable was encoded as 1, 2, 3, and 0 and does not correspond with any documented category. Examination of the UCI repository site confirms that both “ca” and “thal” have missing values in the original dataset. This suggests that the GitHub version replaced those missing values with placeholder integers. This practice could hide data quality issues and potentially introduce errors into the analysis. The official UCI Machine Learning Repository was therefore selected as the data source, accessed using the ucimlrepo Python package with dataset ID 45, to ensure integrity, authenticity, and transparency.

Dataset Description:

The official dataset confirmed the expected missing values: “ca” contained 4 missing values (1.32%), and “thal” contained 2 missing values (0.66%). All other variables were complete. The dataset contains 303 patient records, comprising 13 features and a binary target variable. Of the 13 features, 5 are continuous and 8 are categorically encoded as integers.

The target variable in the original dataset was listed as “num”, and contained 5 levels of disease (0 = no disease, 1 – 4 = varying degrees of disease present). To create a binary target for this project, severity levels 1–4 were collapsed into a single value of 1, representing disease present at any degree. This is consistent with how the Cleveland database has been published in machine learning research. A full data dictionary is provided in Appendix A.

Response Variable:

The response variable is a binary outcome: 0 = No cardiovascular disease present, 1 = Cardiovascular disease present. This outcome was chosen because it directly maps to the clinical decision point — does this patient have CVD requiring further evaluation or intervention? This framing is consistent with how the UCI Cleveland dataset has been used in published literature and is appropriate for a binary classification task.

Data Collection Caveats:

Several important caveats regarding the original data collection methods should be acknowledged. The dataset was collected at the Cleveland Clinic Foundation in the late 1980's, a period when cardiovascular disease was widely considered a male-dominated condition. The dataset is notably male-dominated, with 68% (n=206) male patients compared to 32% (n=97) female patients. It was not until the mid 1980's that researchers began examining how heart disease may affect women (Framingham Heart Disease Study), and at the time, research in men was the gold standard (American Heart Association).

The historical under-representation of women in CVD research has contributed to an incomplete understanding of symptomology and presentation in women (Gauci, et al, 2022). Women are more likely to experience under-represented symptoms leading to potential delays in diagnosis and treatment (Okumura and Murohara, 2025). This means that female patients in the late 1980's may have been less likely to be referred for a diagnostic workup that could have led to inclusion in this dataset. Additionally, the dataset predates modern diagnostic tools including high sensitivity troponin and BNP biomarkers (Wong et al, 2019), and may not reflect current patient populations or contemporary ACC/AHA risk assessment criteria. The goal of this project is not

to deploy a clinical tool but to simulate and demonstrate the clinical validation process that would be applied to one.

DATA CLEANING

Missing Value Assessment

Prior to any preprocessing steps, the dataset was assessed for missing values. A formal Little's MCAR test was implemented via the naniar package in R using the rpy2 bridge in Google Colab. Code attributions were noted: [njtierney/naniar: Tidy data structures, summaries, and visualisations for missing data](#) .

The null hypothesis states that if the data are MCAR (Missing Completely At Random), the p-value is greater than 0.05. The MCAR test returned a p-value of 0.8986, indicating that the null hypothesis is supported, and the missing values in “ca” and “thal” are Missing Completely At Random. The 6 rows with missingness were dropped from both training and test sets without introducing bias. Imputation was considered and deemed clinically inappropriate due to the significance of the values in ca (vessel blockage), and thal (thalassemia type), both of which carry direct clinical meaning that cannot be estimated from other features.

Row Dropping Correction

Initial missing value removal was performed after the train/test split, which resulted in 5 of the 6 rows falling in the test set, creating a disproportionate data loss of 5.9% in the test set compared to 0.5% on the training set. This error was identified and corrected by returning to the original dataset, dropping all 6 rows prior to splitting, and re-splitting using the cleaned version. After

correction, the result was 297 records, split into 213 for training and 84 for testing. Both sets maintained consistent class distributions of 54% no CVD and 46% CVD present, confirming that the stratified split worked correctly.

While Little's MCAR Test confirmed that missingness was completely at random, reducing the likelihood of systematic bias, the small sample size ($n=297$) means that even 6 dropped rows represent ~2% of the data. This represents a minor but acknowledged limitation, given the already small sample size, and should be considered when interpreting the generalizability of the results.

Train/Test Split

The dataset was split into training and test sets prior to any preprocessing steps to prevent data leakage. The split ratio was calculated using a mathematically derived formula based on the number of predictors (Credit: Katherine Geist, PhD, DSE6311). This resulted in a 72/28 training/test split, which gave 213 training records and 84 test records. Stratified splitting was applied and confirmed to maintain the same class distribution as the original dataset.

Encoding

Categorical variables were assessed for encoding requirements prior to modeling. Binary variables (sex, fbs, and exang) were already encoded as 0/1 and did not require further transformation. Ordinal variables, ca and slope, kept their numeric coding as the values carry a meaningful clinical order. One-hot encoding was applied to cp, restecg, and thal using pandas `get_dummies` with `drop_first=True`. Their numeric codes do not hold meaningful weight but do represent clinical categories. Encoding was applied to training and test sets separately to prevent data leakage, and column alignment was used to confirm consistency across both sets. The

variable “thal” was confirmed to be treated nominal rather than ordinal, producing two binary indicator columns – thal_6.0 (Fixed Defect) and thal_7.0 (Reversible Defect)- with thal_3.0 (Normal) as the implicit reference category.

DATA EXPLORATION

Continuous Variables

Descriptive statistics were evaluated for all continuous and categorical variables. The five continuous variables evaluated were age, resting blood pressure (trestbps), serum cholesterol (chol), maximum heart rate achieved (thalach), and ST depression induced by exercise (oldpeak).

Summary statistics are provided in Table 1 and distributional plots in Figure 1(Appendix C).

	Mean	SD	Min	Max	Skewness	Kurtosis
age	54.44	9.04	29.0	77.0	-0.21	-0.52
trestbps	131.69	17.60	94.0	200.0	0.71	0.88
chol	246.69	51.78	126.0	564.0	1.14	4.49
thalach	149.61	22.88	71.0	202.0	-0.54	-0.05
oldpeak	1.04	1.16	0.0	6.2	1.27	1.58

Table 1: Descriptive statistics for continuous variables including mean, standard deviation, skewness, kurtosis, and range.

Age was the most normally distributed continuous variable with a mean of 54.4 years, a range of 29–77, and mild skewness of -0.21, indicating a nearly symmetric distribution. The patient population skews toward middle- to older-age groups, consistent with cardiovascular disease prevalence profiles (CDC, 2023).

Resting blood pressure showed a reasonably normal distribution with a mean of 131.69 mmHg, mild right skew of 0.71, and a maximum value of 200 mmHg consistent with hypertension.

Serum cholesterol showed the most concerning distribution among the continuous variables, with a mean of 246.69 mg/dL, right skew of 1.14, and kurtosis of 4.49 indicating heavy tails and the presence of extreme values. A maximum value of 564 mg/dL was identified as a notable outlier and flagged for preprocessing treatment.

Maximum heart rate achieved during exercise averaged 149.61 bpm with a range of 71-202, and a kurtosis of -0.05. A mild left skew of -0.54 suggests a small concentration of higher heart rate values. ST depression (oldpeak), showed the strongest right skew at 1.27 with kurtosis of 1.58. Most patients clustered near zero, with a long right tail extending to 6.2. This variable may require transformation during preprocessing. Boxplot analysis confirmed outlier patterns suggested in skewness findings. Cholesterol showed the most extreme outlier at 564 mg/dl, sitting well above the upper whisker. Oldpeak showed several outliers as well, ranging from 4.0 to 6.2 (see Figure 2 Appendix C).

Target Variable and Class Distribution

The target variable distribution showed a mild class imbalance with 164 patients without CVD (54.1%), and 139 with CVD (45.9%), as shown in Figure 3. This finding is more balanced than anticipated and unlikely to severely distort model performance. Among the continuous variables, maximum heart rate and ST depression showed the clearest separation between diagnostic classes. Cholesterol and resting blood pressure showed overlap, suggesting weaker predictive value (see Figure 4).

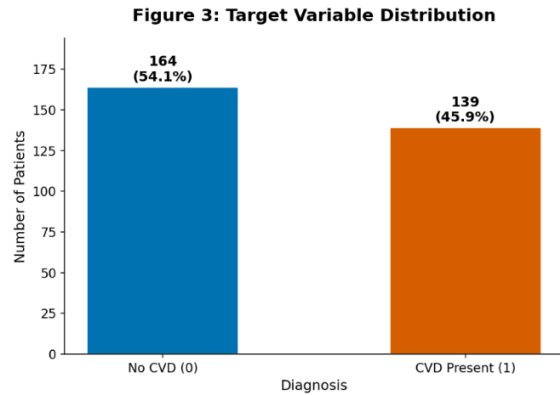


Figure 3: Distribution of binary target variable. Blue = No CVD (n = 164, 54.1%). Orange = CVD present (n = 139, 45.9%). Mild class imbalance shown.

Figure 4: Continuous Variables Stratified by CVD Diagnosis

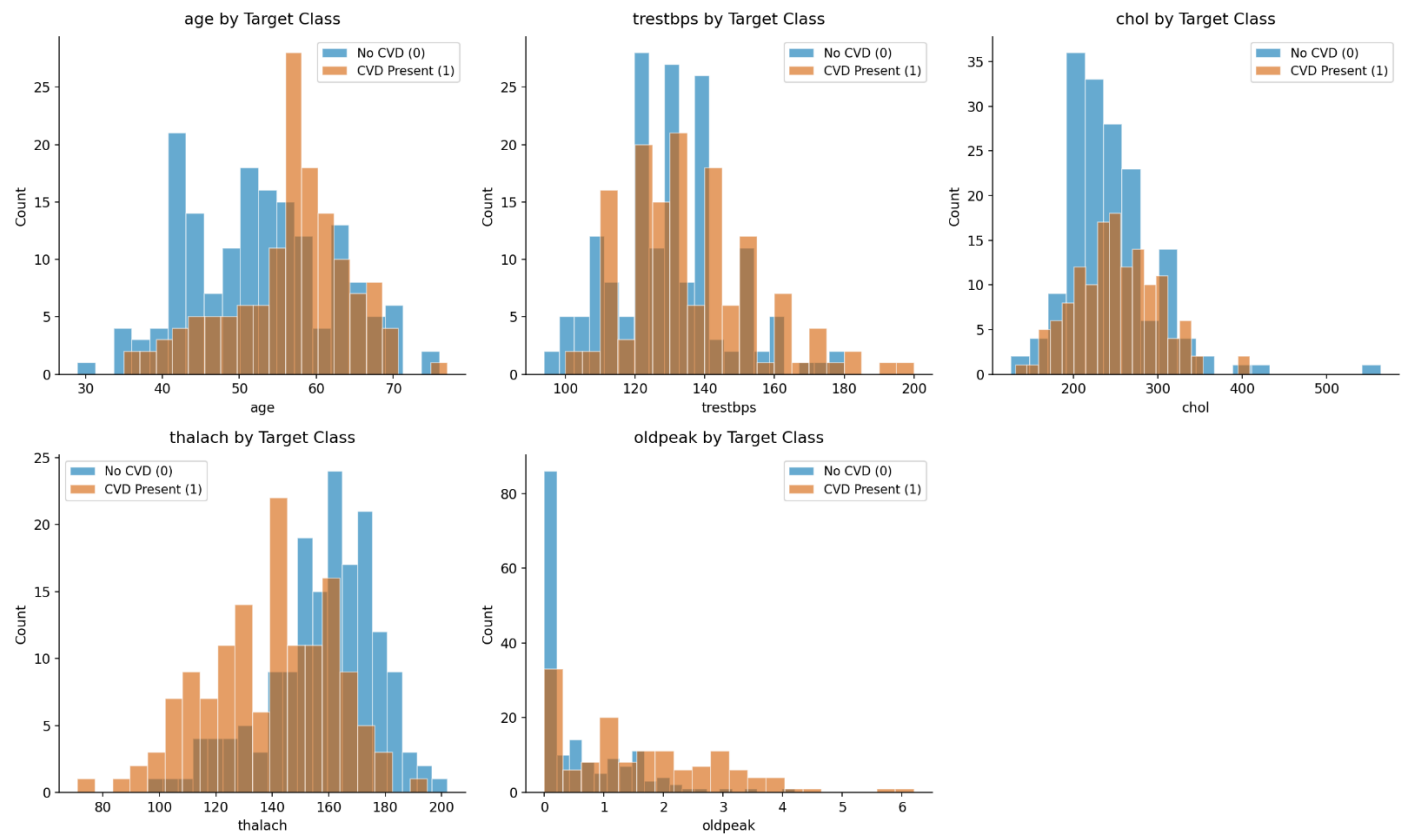


Figure 4: Overlapping histograms of continuous variables stratified by CVD diagnosis. Blue = no CVD. Orange = CVD present. Maximum heart rate (thalach) and ST depression (oldpeak) show clearest separation between classes suggesting a possible stronger predictor value. Cholesterol and resting blood pressure show most overlap between classes.

Categorical Variables

Evaluation of categorical variables revealed clinically meaningful distributions, summarized in Table 2 (see Appendix C). The dataset was notably male-dominated at 68% (n=206), with female patients comprising 32% (n=97), consistent with the data-collection caveats described in the Data Acquisition section.

Asymptomatic chest pain was reported by 47.5% patients, nearly half. This is notable as asymptomatic presentation is a known risk factor for missed cardiovascular disease diagnosis. Fasting blood sugar was elevated in 14.9% of patients. Resting ECG results showed a near even split between normal findings and left ventricular hypertrophy, 49.8% and 48.8%, respectively. Exercise-induced angina was present in 32.7% of patients, representing a significant proportion, suggesting clinical relevance as a marker of coronary artery disease.

When evaluating categorical variables according to CVD diagnosis, the number of major vessels with blockage showed the strongest association with CVD prevalence, increasing across vessel count from 0 to 3. Exercise-induced angina also had a strong CVD prevalence, with over 75% of patients reporting exercise angina having a positive CVD diagnosis.

Asymptomatic chest pain - meaning no noticeable pain is felt, but diagnostic testing such as ECG, imaging or stress tests, confirmed heart disease – showed a high proportion of CVD present. This is consistent with published literature on silent ischemia (Theofilis et al, 2024). These findings will inform feature importance analysis during model development (see Figure 5).

Figure 5: Categorical Variables by CVD Diagnosis (%)

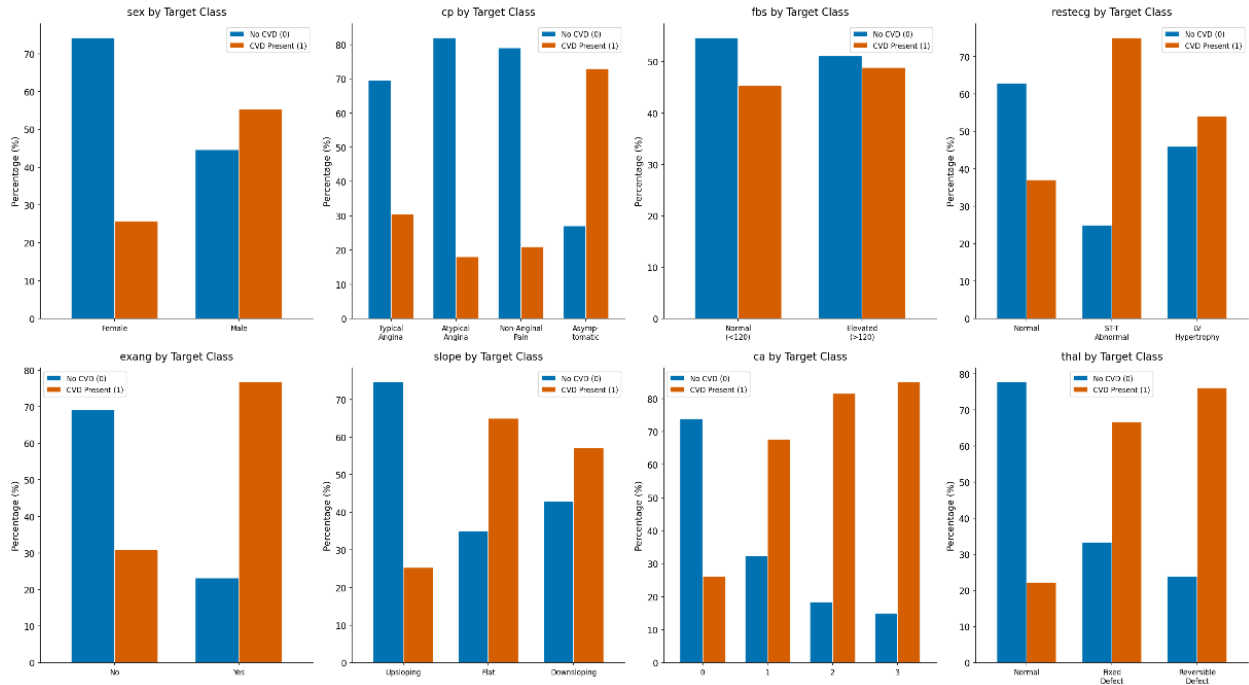


Figure 5: Grouped bar charts showing percentage distribution of CVD diagnosis within each category of 8 categorical predictor variables. Blue = No CVD. Orange = CVD present. Number of blocked vessels (ca), exercise induced angina (Exang), chest pain type (cp), and thalassemia type (thal), show strongest associations with CVD diagnosis.

Correlation Analysis

The correlation heatmap confirmed that no single feature dominates CVD prediction (see Figure 6). Maximum heart rate (thalach) showed the strongest negative correlation with the target variable at -0.42 , meaning higher maximum heart rates are associated with lower CVD prevalence. Thalassemia type (thal), number of blocked vessels (ca), and exercise induced angina (exang) showed the highest positive correlations with CVD presence. Of the predictor variables, oldpeak and slope showed the highest inter-predictor correlation at 0.58 . Both of these variables measure aspects of ST segment behavior and were flagged for multicollinearity during modeling. This finding informed the feature engineering decision described in the Preprocessing section.

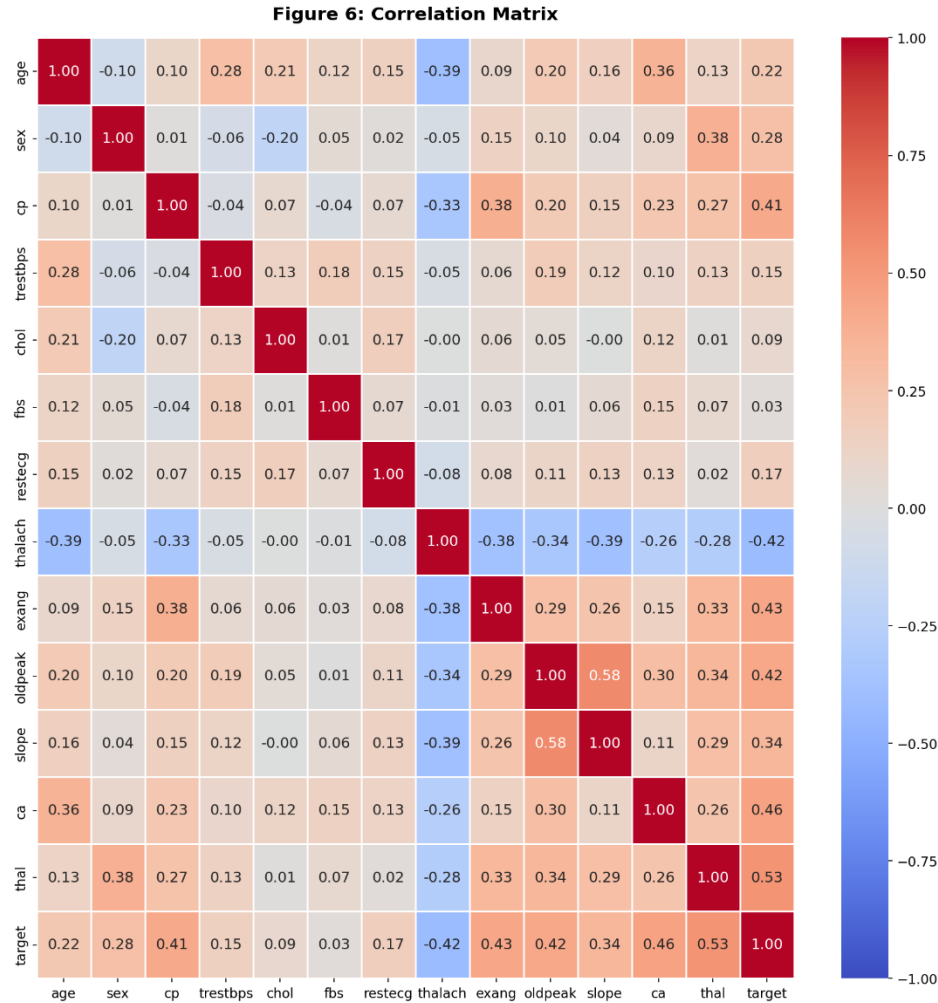


Figure 6: Correlation heatmap of all variables. Red = positive correlation. Blue = negative correlation. Maximum heart rate (thalach), number of blocked vessels (ca), and thalassemia type (thal) show strongest correlations with target variable at -0.42, 0.46, and 0.53, respectively. Oldpeak and slope show the highest possible collinearity at 0.58.

PREPROCESSING AND FEATURE ENGINEERING

Normality Testing and Transformation

Following the train/test split, Shapiro-Wilks tests were run on all continuous variables to assess normality and Q-Q plots were generated to visually confirm findings (see Figure 7). Age and maximum heart rate followed the normal line closely and were considered practically normal. Oldpeak showed the most pronounced departure from normal with points clustering near zero

followed by a sharp rise, consistent with the skewness seen in EDA. Cholesterol showed a mostly normal fit, with an extreme point in the upper tail corresponding to the 564mg/dL outlier.

Based on these findings, Box-Cox transformation was applied to oldpeak and cholesterol, while age, thalach, and trestbps proceeded without transformation.

A shift of +1 was applied to the Box – Cox transformation for oldpeak due to the presence of zero values in that column. Cholesterol responded well and the statistic went from 0.932 → 0.994 and a p-value of 0.000 → 0.505, indicating a successful transformation. Oldpeak statistic showed partial improvement from 0.836 → 0.878, no change in p-value, and remains non-normal due to zero-inflation (see Figure 8).

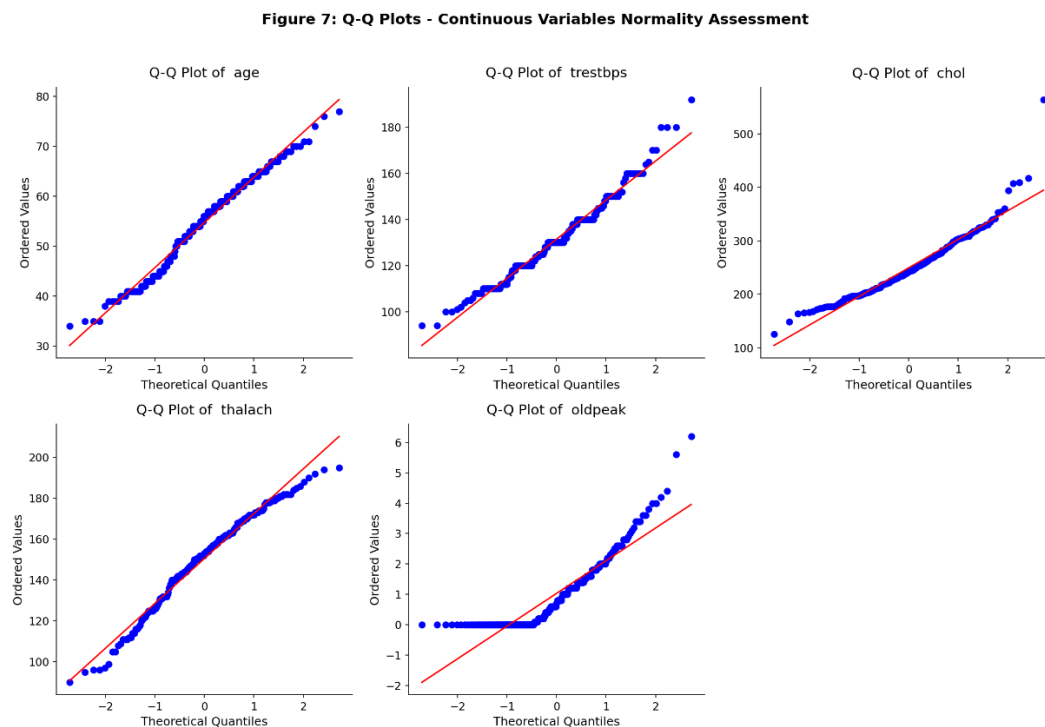


Figure 7: Q-Q plots for five continuous variables. Points on red diagonal indicate normality. Oldpeak shows departure from normal, consistent with right skew in EDA. Cholesterol shows an extreme outlier at upper tail. Age and maximum heart rate show normality despite failing p-value on Shapiro-Wilk test.

Figure 8: Q-Q Plots - Continuous Variables Normality Assessment Post Transformation

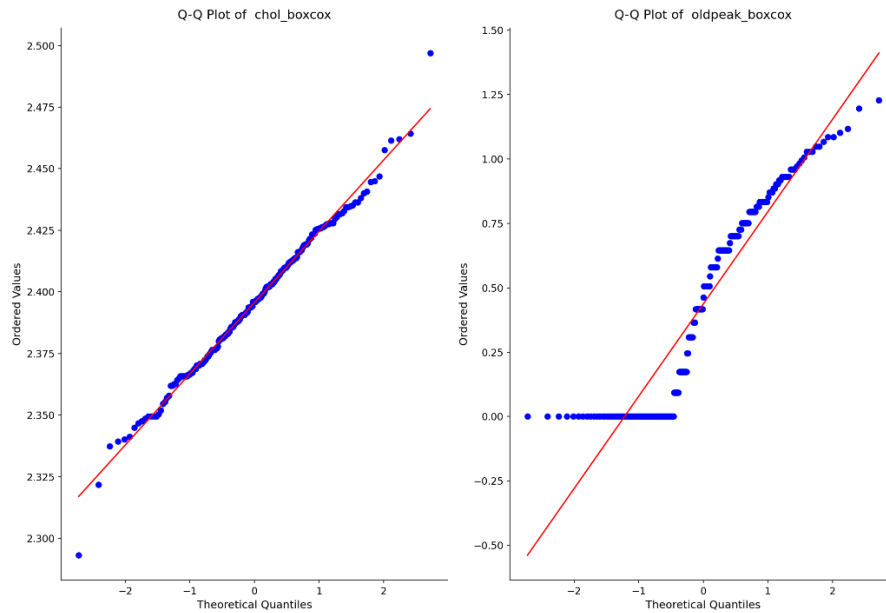


Figure 8: Q-Q plots for cholesterol and oldpeak after Box-Cox transformation. Cholesterol had improved p-value of 0.505 on Shapiro-Wilk. Oldpeak shows improvement but retains zero inflation pattern. Compare with Figure 7.

Feature Engineering

During preprocessing, two related challenges were identified with the ST segmentation variables oldpeak and slope. Oldpeak showed significant zero inflation that Box – Cox transformation improved but did not resolve. Prior EDA identified a moderate correlation between oldpeak and slope ($r=0.58$), suggesting that these variables captured overlapping information. Clinical literature review revealed that these statistical findings reflect a known psychological relationship of ST depression and myocardial ischemia as established in the ACC/AHA Exercise Testing Guidelines (Gibbons et al 1997, 2002). This insight informed a feature engineering decision to combine the two variables into an ordinal severity index with four defined categories: 0 = Negative, 1 = Equivocal, 2 = Positive, 3 = Strongly Positive.

This method resolved both the zero-inflation and multicollinearity issues, while producing a more clinically meaningful feature than two standalone variables. Dimensionality reduction was achieved through clinically informed feature engineering rather than a purely mathematical method while preserving clinical interpretability.

Following the engineering of `st_severity`, a second VIF check confirmed all features remained below the acceptable threshold of 5.0, indicating no introduction of new multicollinearity. Correlation between `st_severity` and the CVD outcome was assessed via crosstabulation, revealing a monotonically increasing CVD presence across severity categories: 31.0% in category 0, 28.3% in category 1, 60.5% in category 2, and 87.2% in category 3. While this strong correlation with the outcome confirms the features predictive ability, it also warrants acknowledgment as a potential source of proxy leakage. `ST_severity` incorporates ST depression and slope measurements collected during stress testing, which may reflect diagnostic findings rather than independent risk factors. This limitation is discussed further in the Discussion section.

Dimensionality Reduction: PCA

Principal Component Analysis was applied to the four continuous variables after standardization to assess whether dimensionality reduction was needed. The variance was relatively evenly distributed across the four principal components. The variables explained 38.6%, 27.7%, 20.4%, and 13.3%, respectively. All four components were required to reach 95% cumulative explained variance, and no clear elbow point was visible on scree plot (see Figure 9). This finding suggests that PCA does not offer meaningful dimensionality reduction for this feature set thus supporting the decision to retain the original engineered features for modeling.

It should be noted that PCA was applied only to continuous variables, as the dataset contains both continuous and categorical variables. A mixed method of unsupervised learning could be applied such as PCA for continuous and a cluster method for categorical, or Factor Analysis of Mixed Data (FAMD), could be more appropriate. Given the time and complexity of project deadlines and criteria, these methods were not explored and represent a potential area for future methodological refinement.

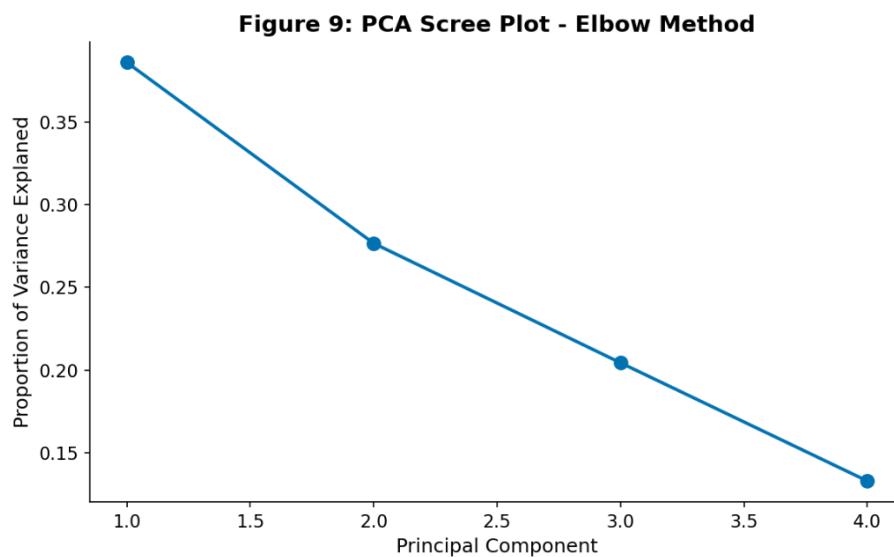


Figure 9: PCA scree plot using elbow method applied to all four continuous variables. Plot shows variance explained per component from PC1 through PC4, 38.6% → 13.3%. The absence of a clear elbow indicates variance is distributed across all four variables and confirms PCA will not reduce feature space.

Scaling

StandardScaler was applied to continuous variables, age, trestbps, chol_boxcox, and thalach. The scaler was fit on the training set only and then applied to both the training and test sets to prevent data leakage. Descriptive statistics confirmed standardization with mean of 0 and standard deviations of 1 on all four variables.

Class Imbalance: SMOTE Exploration

SMOTE was applied to the training set as a proof-of-concept exploration of handling class imbalance. The minority class was oversampled from 98 to 115 records via synthetic patient generation, resulting in a balanced 50/50 split of 230 training records. Due to the mild class imbalance (54%/46%), the impact of SMOTE on model performance was modest, and ROC curves between SMOTE and standard models were nearly identical, suggesting that performance gains were marginal rather than substantial. Due to concerns of synthetic patient data creation in clinical contexts, cost-sensitive learning was ultimately selected as the preferred imbalance handling strategy. CSL addresses class imbalance using real patient data while also embedding clinical risk into the training process by penalizing false negatives more heavily than false positives.

Multicollinearity: VIF

Variance Inflation Factor (VIF) analysis confirmed that any multicollinearity was resolved across the final feature set. All VIF values fell below the 5.0 threshold, with the highest value being sex at 4.01. The reduction of oldpeak and slope into the st_severity engineered feature showed a VIF of 2.65 resolving the two-variable correlation from EDA. The final dataset, with 16 columns, proceeded to modeling with no multicollinearity concerns.

ALGORITHM SELECTION

Logistic Regression: Assumptions and Baseline

Logistic regression was selected as the interpretable clinical baseline model. Three core

assumptions were tested prior to modeling. First, the Box – Tidwell test was applied to assess the linearity of the log-odds assumption for continuous variables. Initial results identified a violation for age ($p=0.001$), while $trestbps(p=0.172)$, $thalach(p=0.848)$, and $chol_boxcox(p=0.078)$ all passed. Age was subsequently binned into clinical categories using scaled threshold equivalents $(-1.04, 0.62)$, and post remediation Box-Tidwell confirmed all remaining continuous variables satisfied linearity: $trestbps(p=0.226)$, $thalach(p=0.785)$, $chol_boxcox(p=0.178)$. Second, VIF confirmed no perfect multicollinearity across the final feature set, as documented in the Preprocessing section. Third, independence of observations was satisfied by data design of one patient per row with no repeated measures.

The standard logistic baseline regression was evaluated using stratified k-fold validation ($k=10$) on the training set. Cross-validation ROC-AUC: 0.887 ± 0.066 , and recall (sensitivity): 0.744 ± 0.154 was below the clinical target of $\geq 85\%$. Train/test gaps were all below 6%, indicating no concerning overfitting. On the held-out test set, the standard model achieved a sensitivity of 74.4% with 10 CVD cases missed, a specificity 91.1%, and an ROC–AUC of 0.932 (see Table 3)(see Figures 10 and 11 in Appendix C).

Figure 12 shows what the model learned and the most meaningful coefficients. Reversible thalassemia defect ($thal_7.0$) was the strongest predictor (coefficient 1.463) – consistent with its known role in CVD. Asymptomatic chest pain (cp_4) was the second strongest predictor, being male (1.018) and number of blocked vessels (ca 0.822) were also strong positive predictors.

Maximum heart rate ($thalach - 0.360$) and non-anginal chest pain ($cp_3 -0.689$) were the strongest negative predictors – higher heart rate associated with absence of CVD, consistent with EDA findings. $St_severity$ (0.410) showed a positive relationship with CVD – validating the feature engineering performed in Preprocessing. Age_binned showed a small coefficient 0.070 –

consistent with a weak age signal and the linearity violation identified during assumption testing. Fasting blood sugar (Fbs) showed minimal predictive contribution (-0.276) – consistent with EDA findings of weak class separation.

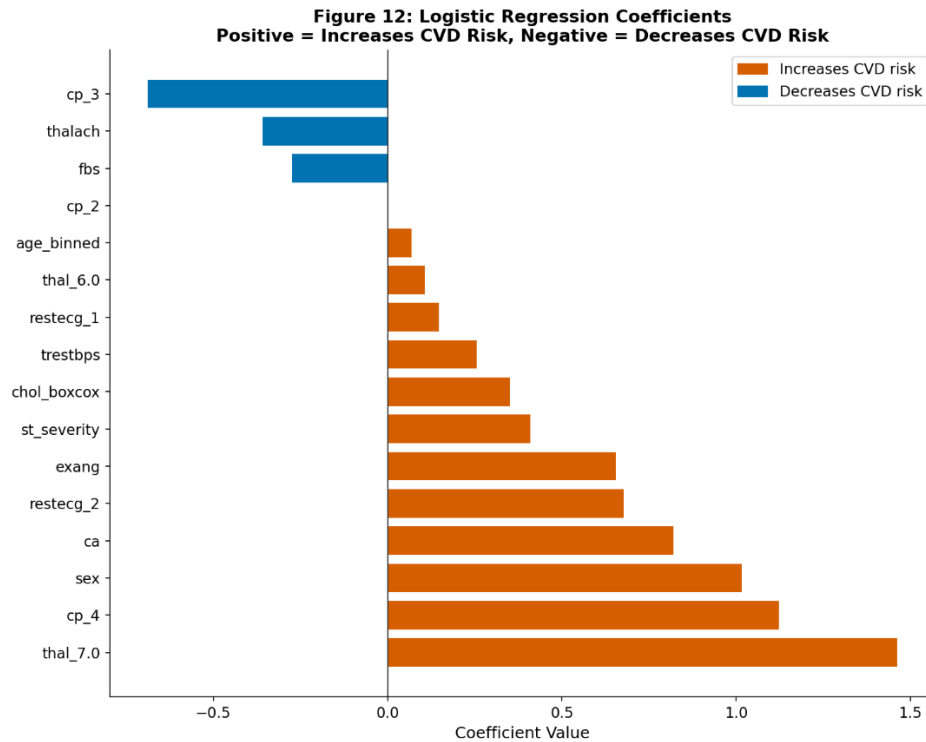


Figure 12: LR coefficient plot.

Logistic Regression: SMOTE and CSL

Three variants of logistic regression were evaluated to assess the effect of class imbalance handling: standard/baseline, SMOTE, and cost-sensitive learning (CSL). SMOTE demonstrated marginal improvement over the standard model. Recall improved from 74.4% → 79.5%, recovering two additional missed CVD cases, while ROC curves between the two models were nearly identical with an AUC difference of only 0.004 (see Figure 14 Appendix). CSL independently produced test set results identical to SMOTE: accuracy 857%, sensitivity 79.5%,

specificity 91.1%, confirming that both approaches address mild class imbalance to a similar degree. Combining SMOTE and CSL produced no additional benefit, as SMOTE's balanced 50/50 split output rendered CSL mathematically redundant. Class weights of 1.0 were assigned to both classes, eliminating any meaningful penalty difference.

CSL was selected as the preferred approach going forward for two reasons. First, CSL uses only real patient data, avoiding the synthetic patient concern associated with SMOTE in clinical contexts. Second, CSL simultaneously addresses class imbalance and embeds clinical risk into the training process by applying a higher misclassification penalty to false negatives, reflecting that a missed CVD diagnosis carries greater patient risk than an unnecessary follow-up. On the original training set, the minority CVD class received a weight of 1.087 compared to 0.926 for the majority class, penalizing missed CVD cases approximately 17% more heavily.

See Figure 16 for ROC-AUC across all LR variants. Confusion matrices for LR SMOTE and LR CSL are provided in the Appendix — results were identical across both variants.

Table 3: Logistic Regression Model Comparison - Test Set Results

	Accuracy	Precision	Recall	Specificity	F1	ROC-AUC	FN	FP
Standard LR	0.833	0.879	0.744	0.911	0.806	0.932	10.0	4.0
SMOTE LR	0.857	0.886	0.795	0.911	0.838	0.936	8.0	4.0
CSL LR	0.857	0.886	0.795	0.911	0.838	0.933	8.0	4.0
SMOTE+CSL LR	0.857	0.886	0.795	0.911	0.838	0.936	8.0	4.0

Table 3: Test set results comparison of all four LR model variants.

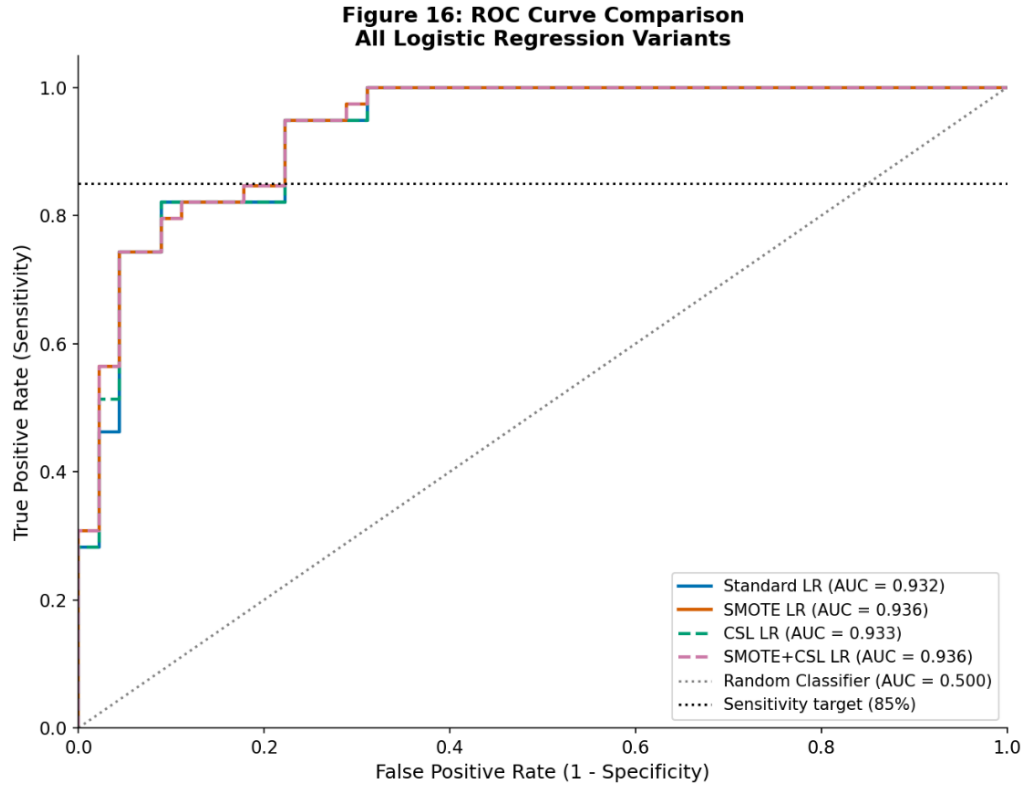


Figure 16: ROC Curve Comparison of all four LR model variants.

Random Forest: Baseline and Tuning

Random Forest was selected as the first ensemble method given its strong performance in comparable CVD classification studies (Vu et al, 2025) and its ability to capture non-linear relationships among predictors. The standard baseline Random Forest showed perfect training scores of 1.000 across all metrics in cross-validation, indicating severe overfitting. Test recall was 0.718 with 11 missed CVD cases both below the clinical target and the LR CSL baseline. Train-test gaps exceeded 10% across all metrics, confirming that hyperparameter tuning was required before any meaningful interpretation. Applying CSL to the untuned baseline worsened performance further. Sensitivity dropped to 66.7% with 13 missed CVD cases.

Hyperparameter tuning was performed using GridSearchCV with RepeatedStratifiedKFold (k=10, repeats=2) and recall as the scoring metric, consistent with the clinical sensitivity priority. This approach ensured that tuning decisions were consistent with the threshold-based evaluation, and models optimized for recall during tuning were then assessed at adjusted thresholds to confirm whether clinical targets could be met in practice.

GridSearchCV parameter grids:
Random Forest - Hyperparameter Tuning

n_estimators': [100, 200]
max_depth: [3, 5, 10]
min_samples_leaf: [2, 4]
class_weight: ['balanced', None]

Best parameters selected were max_depth=3, min_samples_leaf=4, n_estimators=100, and class_weight='balanced'. Tuning successfully resolved severe overfitting, train/test gaps were reduced to under 8% across all metrics and recall improved 0.724 → 0.756 in cross-validation. On the held-out test set, RF Tuned achieved an accuracy of 82.1%, a recall of 74.4%, a specificity of 88.9%, and an ROC-AUC of 0.928, with 10 missed CVD cases, which is comparable to the LR standard baseline but not better than the LR CSL on any metric.

XGBOOST: Baseline and Tuning

XGBoost was selected as the second ensemble method due to its gradient-boosting architecture and strong performance in the published CVD prediction literature. Like Random Forest, the standard XGBoost baseline showed severe overfitting with perfect training scores and a test recall of 74.4% with 10 missed CVD cases. XGB CSL Baseline performed worse than XGB Standard on all metrics before tuning – sensitivity dropped from 74.4% to 66.7%, confirming the

same interaction between overfitting and class weighting observed in Random Forest.

Two rounds of hyperparameter tuning were required to adequately address overfitting. Round 1 focused on structural parameters:

XGBoost Round 1

n_estimators: [100, 200]

max_depth: [3, 5, 7]

learning_rate: [0.01, 0.1, 0.3]

subsample: [0.8, 1.0]

scale_pos_weight: [1, spw]

The best parameters were n_estimators=100, max_depth=3, learning_rate=0.1, and subsample=0.8.

Round 2 focused on regularization:

XGBoost Round 2

min_child_weight: [1, 3, 5]

gamma: [0, 0.1, 0.3]

reg_alpha: [0, 0.1, 0.5]

reg_lambda: [1, 1.5, 2]

The best parameters found were gamma=0.3, min_child_weight=5, reg_alpha=0.1, and reg_lambda=2. Following two rounds of tuning, the train-test gap reduced to 0.054 and cross-validation recall improved to 0.817. On the held-out test set, XGB Tuned R2 achieved recall of 74.4% and ROC-AUC of 0.917 showing strong discriminative ability but 10 missed CVD cases, matching RF Tuned and falling short of LR CSL. The discrepancy between cross-validation

recall (0.817) and test-set recall (0.744) is attributed to the small test set size (n=39 CVD patients), in which individual patient classifications have an outsized effect on aggregate metrics.

SVM: Baseline and Tuning

Support Vector Machine was added as a fourth model, given its known suitability for small datasets. SVM's margin maximization provides built-in regularization which makes it less prone to overfitting on limited sample sizes. Both linear and RBF kernels were evaluated. The `probability=True` parameter was set for all SVM models, enabling Platt scaling for probability output, which is required for threshold and Decision Curve Analysis.

The SVM linear baseline achieved a test sensitivity of 76.9% with 9 missed CVD cases, specificity at 93.3% and ROC-AUC of 0.946, the highest of any baseline model. The training data in cross-validation showed some minor overfitting, which was addressed using `GridSearchCV` for hyperparameter tuning. `GridSearchCV` with `RepeatedStratifiedKFold` (k=10, repeats=2) was applied to tune the regularization parameter C across values of 0.01, 0.1, 1, 10, and 100. The optimal value of C=1 (default value) was selected, indicating that the data did not benefit from aggressive regularization in either direction.

After tuning, SVM improved recall from 0.769 to 0.795 with 8 missed CVD cases, specificity of 91.1%, and ROC- AUC at 0.944

The RBF kernel was evaluated at baseline and produced identical test set classifications to the linear kernel with marginally lower AUC (0.940 vs 0.946). RBF tuning was omitted since it showed no performance benefit with added complexity. This decision was made on the principle that the simplest model with equal results is preferred over a more complex one.

Overfitting Control

Overfitting was controlled through four complementary strategies across all models. First, stratified k-fold cross-validation(k=10) was applied to all models during evaluation.

RepeatedStratifiedKFold (k=10, repeats=2) was used within GridSearchCV for hyperparameter tuning to ensure class distribution was maintained across all folds. Second, the train/test gap was monitored across all metrics. Gaps exceeding 10% were flagged as severe overfitting requiring tuning intervention. Third, regularization was applied through model-specific mechanisms: L2 penalty for logistic regression, max_depth and min_samples_leaf for Random Forest, gamma and reg_lambda for XGBoost, and C parameter for SVM. Fourth, the held-out test set was used only once for final evaluation after all tuning decisions were made to ensure an unbiased estimate of performance.

FINAL/BEST MODEL

Model Selection Justification

After evaluation of all models at baseline and after tuning, Logistic Regression with Cost-Sensitive Learning (LR CSL) was selected as the final best model. The full model comparison across all baseline and tuned variants is provided in Table 4 and Figure 17. At the default classification threshold of 0.50, LR CSL achieved an accuracy of 857%, a recall of 79.5%, a specificity of 91.1%, an F1 of 0.838, and an ROC-AUC of 0.933. This model performed better than all tuned ensemble methods on recall and FN count with 8 missed CVD cases.

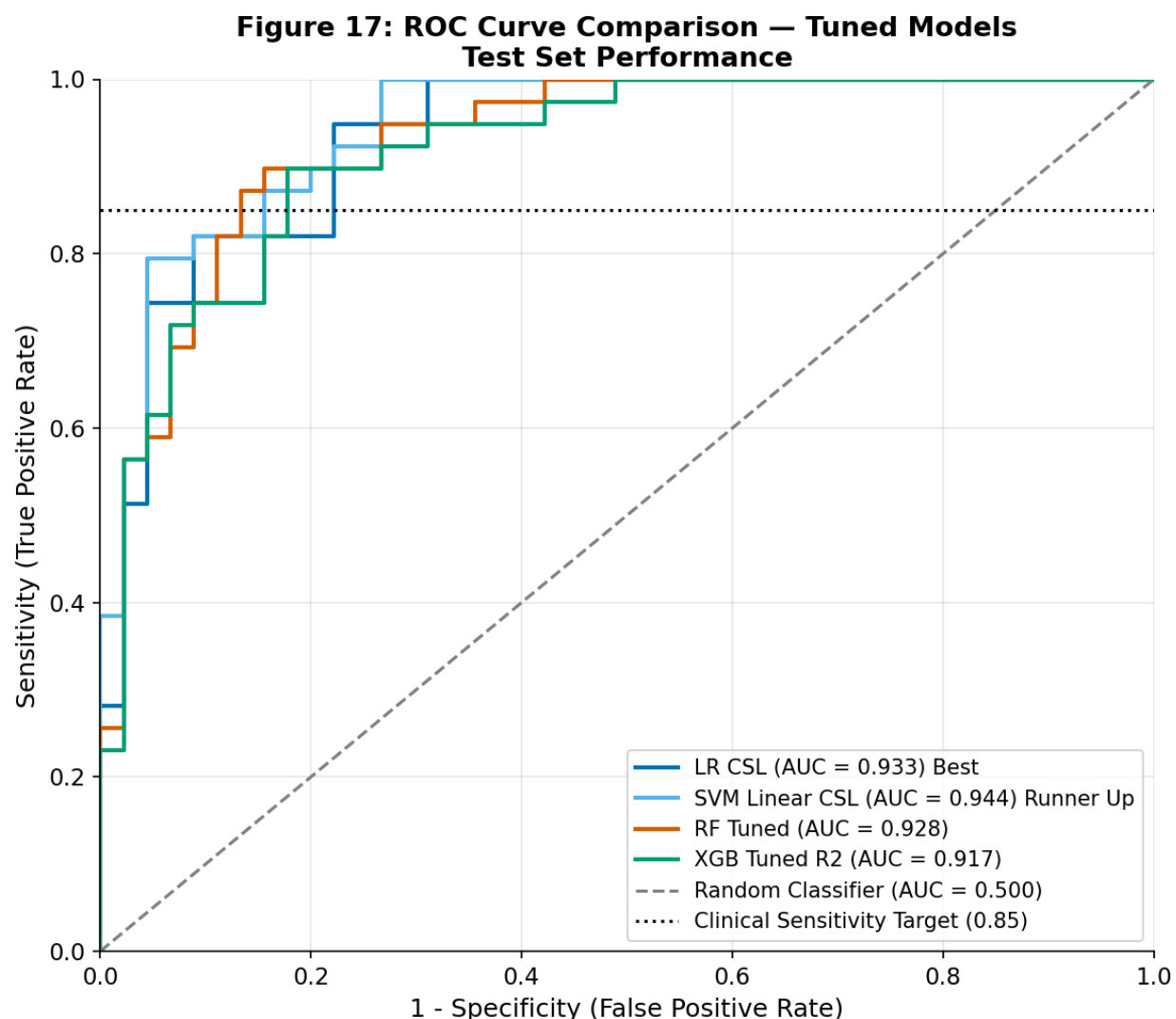


Figure 17: ROC curve comparison for all four tuned models on held-out test set. Dotted line at sensitivity=0.85 indicates clinical sensitivity target. All models cross the target line at approximately FPR=0.20, confirming threshold optimization is achievable.

Although SVM Linear CSL demonstrated a marginally higher AUC (0.944 vs 0.933), threshold analysis showed that LR CSL achieved higher sensitivity at the optimized threshold of 0.25, missing only 2 CVD cases, compared to 3 for SVM Linear CSL. The reduction in false negatives outweighs a marginal AUC difference that is unlikely to be statistically significant on a test set of 84 patients in a clinical context. Also, logistic regression produces directly interpretable

coefficients showing how each feature influences the prediction. LR CSL was therefore selected on both clinical utility and methodological transparency.

Table 4: Full Model Comparison - Baselines vs Tuned Models - Test Set Results

	Accuracy	Recall	Specificity	F1	AUC	FN	FP
LR Standard	0.833	0.744	0.911	0.806	0.932	10	4
RF Standard	0.810	0.718	0.889	0.778	0.909	11	5
XGB Standard	0.821	0.744	0.889	0.795	0.882	10	5
SVM Linear	0.857	0.769	0.933	0.833	0.946	9	3
SVM RBF	0.857	0.769	0.933	0.833	0.940	9	3
LR CSL	0.857	0.795	0.911	0.838	0.933	8	4
RF Tuned	0.821	0.744	0.889	0.795	0.928	10	5
XGB Tuned R2	0.821	0.744	0.889	0.795	0.917	10	5
SVM Linear CSL	0.857	0.795	0.911	0.838	0.944	8	4

Table 4: Full model comparison across all baseline and tuned models. Test set results including accuracy, recall, specificity, F1, ROC-AUC, false negatives, and false positives. Green row indicates selected best model (LR CSL). Blue row indicates runner up (SVM Linear CSL).

The finding that LR CSL outperformed tuned ensemble methods is consistent with published literature demonstrating that logistic regression performs comparably or better than machine learning approaches for predicting major chronic diseases on small clinical datasets (Nusinovici et al, 2020; Couronne et al, 2018). Model complexity does not necessarily translate into improved clinical prediction performance which is a finding that directly supports the core thesis of this project.

SHAP Feature Importance

SHAP (Shapley Additive exPlanations) analysis was applied to the LR CSL model to interpret feature contributions to CVD predictions (see Figure 18). SHAP analysis identified `thal_7.0` (reversible thalassemia defect), `ca` (number of vessels blocked), and `cp_4` (asymptomatic chest

pain) as strongest predictors of CVD presence. The engineered st_severity ranked 5th overall, validating the feature engineering decision made during preprocessing. Features at the bottom of the importance list include restecg_1, thal_6.0, and cp_2, and may warrant removal in future iterations with larger datasets.

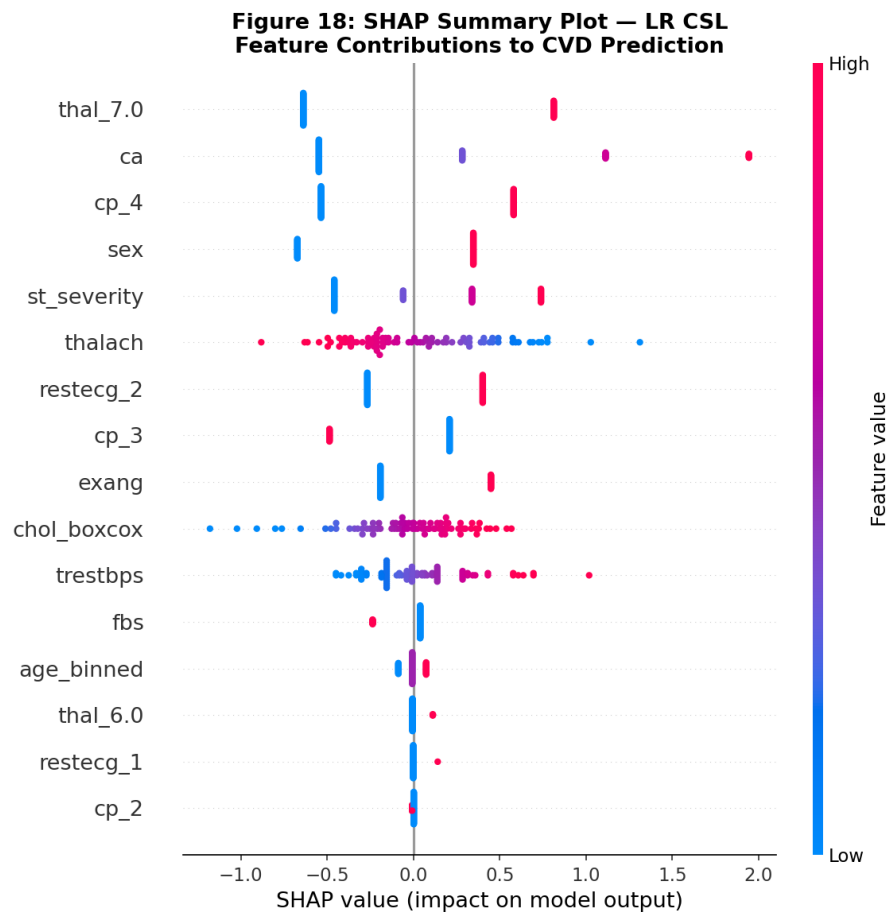


Figure 18. SHAP summary plot for LR CSL model on held-out test set. Features ranked top to bottom by overall importance. Dot position on x-axis indicates direction and magnitude of contribution — right indicates push toward CVD, left indicates push away from CVD. Color indicates feature value magnitude — red=high, blue=low.

Threshold Analysis

At the default classification threshold of 0.50, no model achieved the clinical sensitivity target of ≥ 0.85 . This highlights the inadequacy of statistical defaults in clinical contexts. Threshold analysis tested every cutoff from 0.10 to 0.90 across all four tuned models and calculating

sensitivity and specificity at each point to identify where both targets could be met simultaneously (see Figure 19).

All four tuned models demonstrated the capability to meet both targets with threshold optimization. LR CSL achieved the highest sensitivity of 0.949 at a threshold of 0.25, missing only 2 CVD cases, and a specificity of 0.778. SVM Linear CSL achieved a sensitivity of 0.923 at a threshold of 0.30 with 3 missed cases. RF Tuned showed the best sensitivity/specificity balance at thresholds 0.40, 0.897, and 0.844, respectively, and generated only 7 false alarms, compared to 10 for LR CSL. In a resource-constrained clinical environment where unnecessary referrals and tests incur operational costs, RF Tuned at a threshold of 0.40 may be a preferable trade-off. The selection of the final deployment model is a clinical governance decision requiring input from cardiologists and hospital administrators.

Cost-sensitive learning and threshold optimization are complementary but distinct mechanisms. CSL shifted the model's decision boundary during training by weighting the CVD class at 1.087 vs 0.926 for the no-CVD class. This penalized missed diagnoses ~17% more heavily and improved recall at the default threshold from 0.744 to 0.795. Threshold optimization then applied a post-hoc clinical decision rule, lowering the cutoff from 0.50 to 0.25 to flag any patient with predicted CVD probability above 25%. These two mechanisms together produced a sensitivity of 0.949, catching 37 of 39 CVD cases, and maintaining a specificity of 0.778. CSL embeds clinical risk into the model's learned parameters and threshold adjustment applies it to the model's probability outputs.

**Figure 19: Threshold Analysis — Sensitivity & Specificity Across Thresholds
Tuned Models — Test Set**

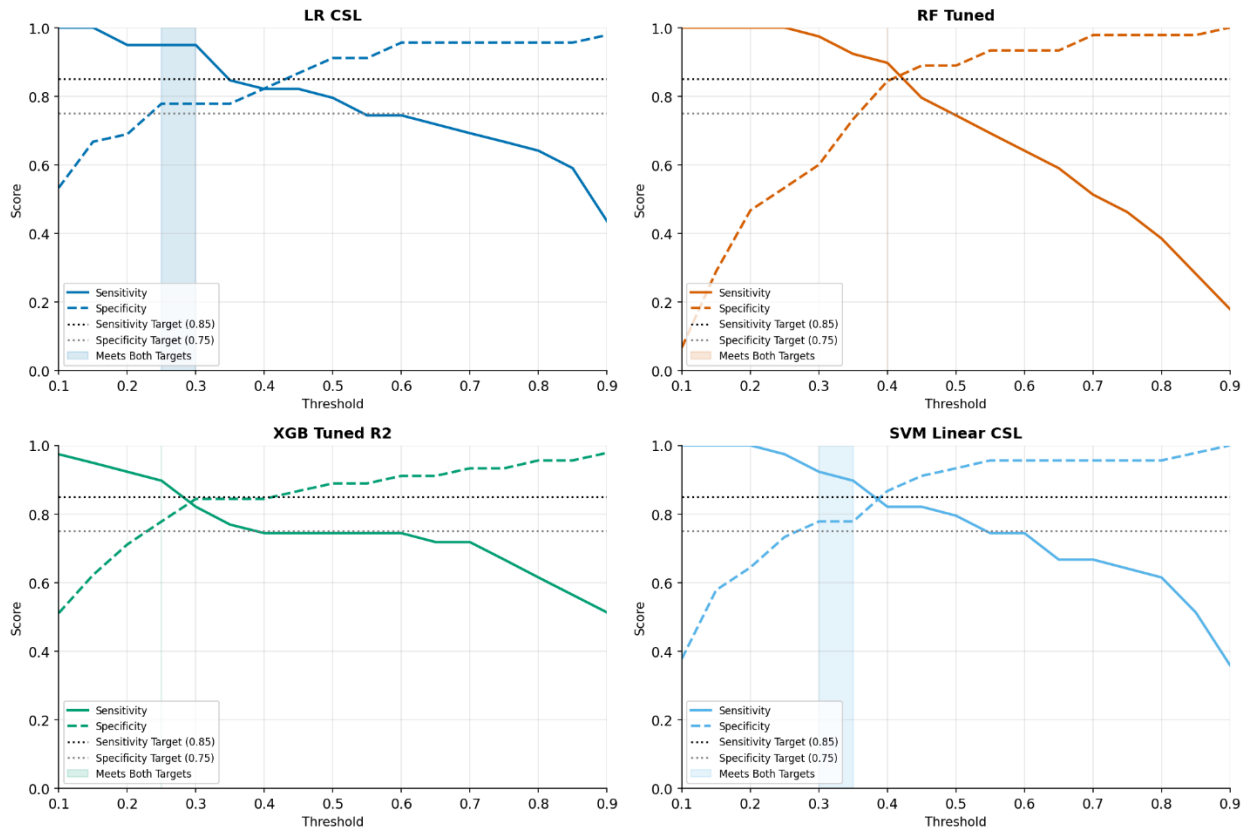


Figure 19: Threshold analysis for all four tuned models. Solid lines represent sensitivity, dashed lines represent specificity across thresholds 0.10–0.90. Dotted black line indicates sensitivity target (0.85), dotted grey line indicates specificity target (0.75). Shaded regions indicate threshold range where both clinical targets are met simultaneously.

Decision Curve Analysis

Decision Curve Analysis was applied to confirm clinical use in all four tuned models (see Figure 20). DCA evaluates whether acting on model predictions produces better patient outcomes than two alternatives: treating all patients or treating no patients. The treat-none baseline represents a net benefit of zero, while the treat-all baseline declines as threshold probability increases due to the growing penalty for unnecessary interventions.

All four model curves showed net benefit above both baselines across the relevant threshold range of 0.10 – 0.40. LR CSL and SVM Linear CSL show the highest net benefit, further supporting LR CSL as the recommended model for clinical deployment. These results show that using the model to guide decision making would be better than treating all or treating no patients.

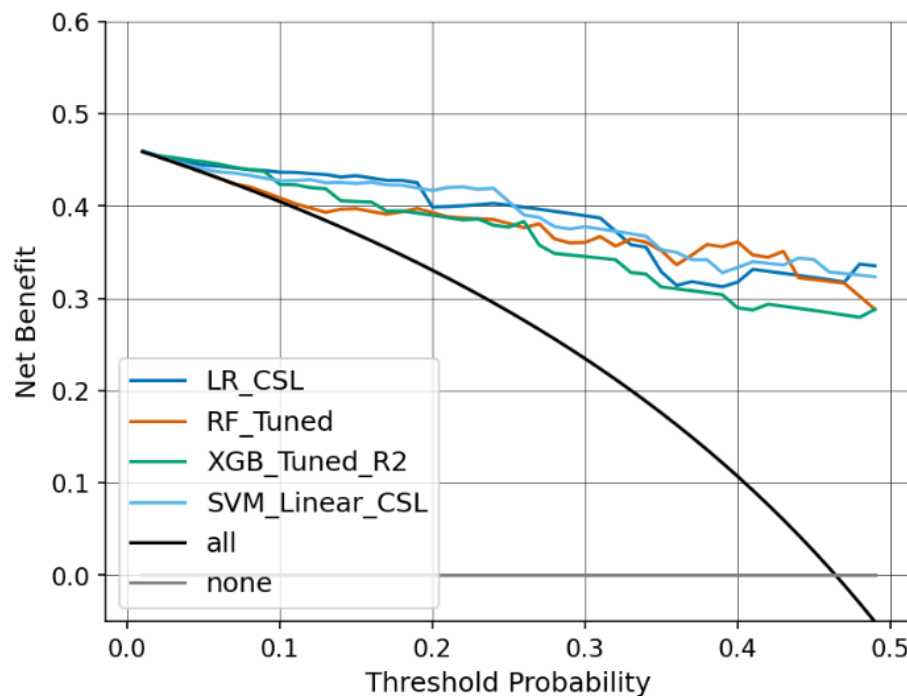


Figure 20: Decision Curve Analysis for all four tuned models. Model curves above treat-all (black) and treat-none (grey at zero) baselines confirm clinical utility across threshold range 0.10–0.40.

Proxy Leakage

Several top predictors, including *thal*, *ca*, *exang*, *restecg*, and *st_severity*, are diagnostic findings measured during the cardiovascular workup rather than pre-existing risk factors. This represents a form of proxy leakage, meaning that the model is partly learning from findings discovered during the diagnostic process rather than from independent predictors. This does not invalidate the model but scopes it as a diagnostic classification tool for patients already undergoing cardiovascular workup, to include stress testing and coronary angiography, rather than a

prospective population-level risk prediction tool. This limitation is consistent with how the UCI Cleveland dataset has been used in published literature and will be acknowledged in the Discussion section.

DISCUSSION & NEXT STEPS

Key Takeaways

The project set out to answer a core question: Is accuracy enough to validate a clinical machine learning model? The answer, derived from a three-layer validation framework, is no. Standard accuracy metrics alone do not capture the sensitivity and specificity tradeoffs required for clinically appropriate model evaluation. A model achieving 85.7% accuracy still missed 8 CVD patients at the default threshold. That finding would not be visible in an accuracy-only framework.

The initial prediction that ensemble methods would outperform Logistic Regression was not supported. LR CSL emerged as the best performing model which is consistent with Couronne et al 2018 in that Logistic Regression outperforms complex ensemble methods on small clinical datasets. Aside from dataset size, the failure of ensemble methods can be attributed to three factors: 1) Random Forest and XGBoost are high-variance models that require large sample sizes to average out individual tree instability. With only 213 training records, the variance reduction benefit of ensembling was insufficient to overcome this limitation. 2) The feature space of the dataset is largely linearly separable, as was confirmed by near identical performance of linear SVM and logistic regression. Non-linear decision boundaries offered by ensemble methods provide no additional discriminative benefit. 3) The feature engineering performed during preprocessing, specifically `st_severity`, resolved the primary sources of non-linearity and

multicollinearity in the data. This provided a feature set well-suited to linear classification.

Together these factors created conditions where a well-regularized linear model was the optimal choice, not a limitation of the analysis.

The core elements of the hypothesis were validated: Cost-sensitive learning improved performance across all models, and threshold optimization was confirmed as essential for meeting clinical targets that default settings could not achieve. By lowering the threshold from 0.50 to 0.25, six additional CVD cases were recovered while increasing false positives from 4 to 10, a tradeoff with real consequences for patients and healthcare resources.

Evaluated at an optimized threshold of 0.25, LR CSL achieved a sensitivity 94.9% of true CVD cases, while maintaining specificity above the clinical minimum of 0.75. Decision Curve Analysis confirmed clinical usefulness across the relevant threshold range, providing evidence that model deployment would produce better patient outcomes than treating all or none of the patients. These results are consistent with findings from comparable CVD classification studies achieving a sensitivity of 89.01% and a specificity of 85.39% using similar methodological approaches (Vyshynya et al, 2024)

Threats to Validity

Several internal and external threats to validity must be acknowledged. Internally, the small sample size of $n=297$ limits statistical power and increases the risk that results are sample-specific patterns rather than general relationships. The train/test split of 213/84 means the held-out test set contains 39 CVD-positive patients. Individual patient classifications have an outsized effect on aggregate metrics and small differences in false negative count (eg. 2 vs 3 missed

cases) may not be statistically meaningful at this sample size. Stratified k-fold cross-validation was applied throughout to mitigate this concern, but it cannot fully compensate for limited data.

Externally, the dataset was collected at a single institution in 1988, limiting generalizability to modern and diverse patient populations. The male-dominated composition (68%) means model performance for female patients is less well-characterized, and with known differences in CVD symptoms between sexes, results should be interpreted with caution for female populations (Gauci et al, 2022; Okumura et al, 2025). Also, the dataset predates modern diagnostic tools such as high-sensitivity troponin and BNP biomarkers, as well as contemporary ACC/AHA risk assessment criteria. Additionally, patients in this dataset were referred for cardiac evaluation at the Cleveland Clinic Foundation, representing a higher-risk population rather than a general population sample, which limits generalizability to broader screening contexts.

Proxy leakage represents an additional internal threat. The features `ca`, `thal`, `exang`, `restecg`, and `st_severity`, are measured during or after the diagnostic workup rather than prior to it. This classifies the model as a diagnostic tool for patients already undergoing cardiovascular evaluation, rather than a prospective risk predictor for the general population. Deploying this model outside of cardiac stress testing would be methodologically inappropriate and possibly unsafe.

Sources of Bias

Several potential sources of bias were identified that could not be fully accounted for within the scope of this project. The dataset inherently suffers from selection bias. The patients included were those referred for cardiac evaluation and diagnostic workup, representing a higher-risk referred subpopulation rather than a general population sample. Demographic bias is present due

to the historical under-representation of women and the single institution collection environment, which may not reflect the diversity of modern patient populations. A temporal bias exists due to the dataset being collected in 1988. Cardiovascular disease risk profiles, diagnostic criteria, and patient demographics have changed substantially in the subsequent decades.

Dream Dataset

The limitations of the UCI Heart Disease dataset highlight what an ideal dataset for this type of clinical validation study would include. A dream dataset for CVD classification validation would include a substantially larger sample ($n > 10000$) from multiple institutions across diverse geographic and demographic populations to support meaningful generalization. It would include modern biomarkers such as high-sensitivity troponin, B-type natriuretic peptide (BNP), coronary artery calcium scoring, advanced findings from coronary CTA, and genetic risk markers.

Longitudinal follow-up data linking diagnostic findings to patient outcomes would enable prospective risk prediction rather than retrospective classification. Contemporary ACC/AHA risk variables and standardized electronic health record data fields would ensure compatibility with clinical workflows. This type of dataset would substantially strengthen the clinical validation framework demonstrated in this project and support the development of a model appropriate for prospective deployment consideration.

Recommendations

For clinical AI and machine learning developers, this project demonstrates that a three-layer validation framework, including standard metrics, threshold analysis, and Decision Curve Analysis, should be the minimum evaluation standard for any clinical classification model. Accuracy and AUC alone are insufficient. Threshold selection should be treated as a clinical governance decision involving cardiologists, hospital administrators, and patient safety teams.

External validation on a larger and contemporary dataset is the immediate recommended next step before any consideration of clinical deployment. Pursuit of FDA clearance as a clinical decision support tool would require institutional review board approval, prospective clinical validation, and regulatory consultation.

For cardiologists and clinical informaticists, the recommended deployment threshold of 0.25 would flag any patient with a predicted CVD probability of 25% or higher for further clinical evaluation. At this threshold, 94.9% of true CVD cases are caught with 10 false positives per 84 patients. In settings where resources are tight and unnecessary referrals carry operational costs, the RF Tuned model at threshold 0.40 offers a sensitivity/specificity balance with 7 false positives. The final threshold selection should reflect the specific clinical context, patient population, and resource constraints of the deployment environment. This model should not replace informed clinical judgment but should support it.

Future Directions

Several directions for future work were identified during this project. The most important is external validation on a larger and more contemporary dataset is required before any deployment consideration. Next, a feature set should be expanded to include modern biomarkers and imaging findings that better reflect current clinical practice. The proxy leakage limitation could be addressed by separating pre-test risk factors from diagnostic findings and building a two-stage model, one for prospective screening and one for diagnostic classification, each validated independently. The application of Factor Analysis of Mixed Data (FAMD) as an alternative to PCA for dimensionality reduction on mixed and continuous datasets could be explored in future iterations. A Delong's test should be applied to formally assess whether the AUC difference between LR CSL and SVM Linear CSL is statistically significant; this could resolve the model

tiebreaker on statistical grounds and not just clinical. Finally, integration with a clinical workflow, including EMR compatibility assessment and clinician adoption evaluation, would be required to complete the full clinical AI validation lifecycle demonstrated in this project.

CODE AVAILABILITY

All code, data, and analysis notebooks for this project are publicly available on GitHub. The repository includes cleaned datasets, preprocessing pipelines, model training and evaluation notebooks, and the final report PDF. The repository can be accessed at:

https://github.com/mnooney/CVD_Capstone

Appendix A:

Data Dictionary

Variable	Type	Description	Values/Units
age	Continuous	Patient age	Years
sex	Binary	Biological sex	0=Female, 1=Male
cp	Nominal	Chest pain type	1=Typical angina, 2=Atypical angina, 3=Non-anginal pain, 4=Asymptomatic
trestbps	Continuous	Resting blood pressure	mmHg
chol	Continuous	Serum cholesterol	mg/dL
fb	Binary	Fasting blood sugar >120 mg/dL	0=False, 1=True
restecg	Nominal	Resting ECG results	0=Normal, 1=ST-T wave abnormality, 2=Left ventricular hypertrophy
thalach	Continuous	Maximum heart rate achieved	bpm
exang	Binary	Exercise induced angina	0=No, 1=Yes
oldpeak	Continuous	ST depression induced by exercise	mm
slope	Ordinal	Slope of peak exercise ST segment	1=Upsloping, 2=Flat, 3=Downsloping
ca	Ordinal	Number of major vessels colored by fluoroscopy	0-3
thal	Nominal	Thalassemia type	3=Normal, 6=Fixed defect, 7=Reversible defect
target	Binary	CVD diagnosis	0=No CVD, 1=CVD Present
Engineered features:			
Variable	Type	Description	Values
chol_boxcox	Continuous	Box-Cox transformed serum cholesterol	Transformed units
st_severity	Ordinal	Engineered ST severity score combining oldpeak and slope per ACC/AHA guidelines	0=Negative, 1=Equivocal, 2=Positive, 3=Strongly Positive
age_binned	Ordinal	Age binned into clinical categories using scaled threshold equivalents	0=<45 years, 1=45-60 years, 2=>60 years

Appendix B:

Abandoned Methods

Several analytical approaches were considered and ruled out during this project. The following is documented for transparency and replication purposes.

Random Oversampling was considered an alternative to SMOTE for handling class imbalance. It was rejected due to the risk of overfitting from exact record duplication. Repeating identical patient records in the training set inflates model performance without adding new information.

SMOTE (Synthetic Minority Oversampling Technique) was explored as a proof-of-concept and produced marginal performance improvements over the standard model. It was ruled out as the primary imbalance-handling strategy due to concerns about synthetic patients in clinical contexts. The interpolated data points between real patient records may not reflect clinically valid presentations.

SMOTE + CSL combination was tested and confirmed redundant. SMOTE's balanced 50/50 output resulted in CSL being mathematically redundant and producing class weights of 1.0 for both classes with no meaningful penalty difference. CSL alone on the training data was selected as the preferred approach.

PCA (Principal Component Analysis) was assessed for dimensionality reduction on the four continuous variables. The variance was evenly distributed across all four components with no clear elbow point in the scree plot, confirming PCA would not meaningfully reduce dimensionality. Clinically informed feature engineering via `st_severity` was used instead.

SVM RBF kernel was evaluated at baseline and produced identical test set classifications to the linear kernel with marginally lower AUC and larger cross-validation overfitting gaps. RBF tuning was omitted on the principle that the simplest model producing equivalent results is preferred over a more complex one.

FAMD (Factor Analysis of Mixed Data) was identified as a more methodologically appropriate alternative to PCA for datasets containing both continuous and categorical variables. It was not implemented due to project scope and time constraints but is recommended for future iterations.

Appendix C:

Additional Figures

Figure 1: Distribution of Continuous Variables

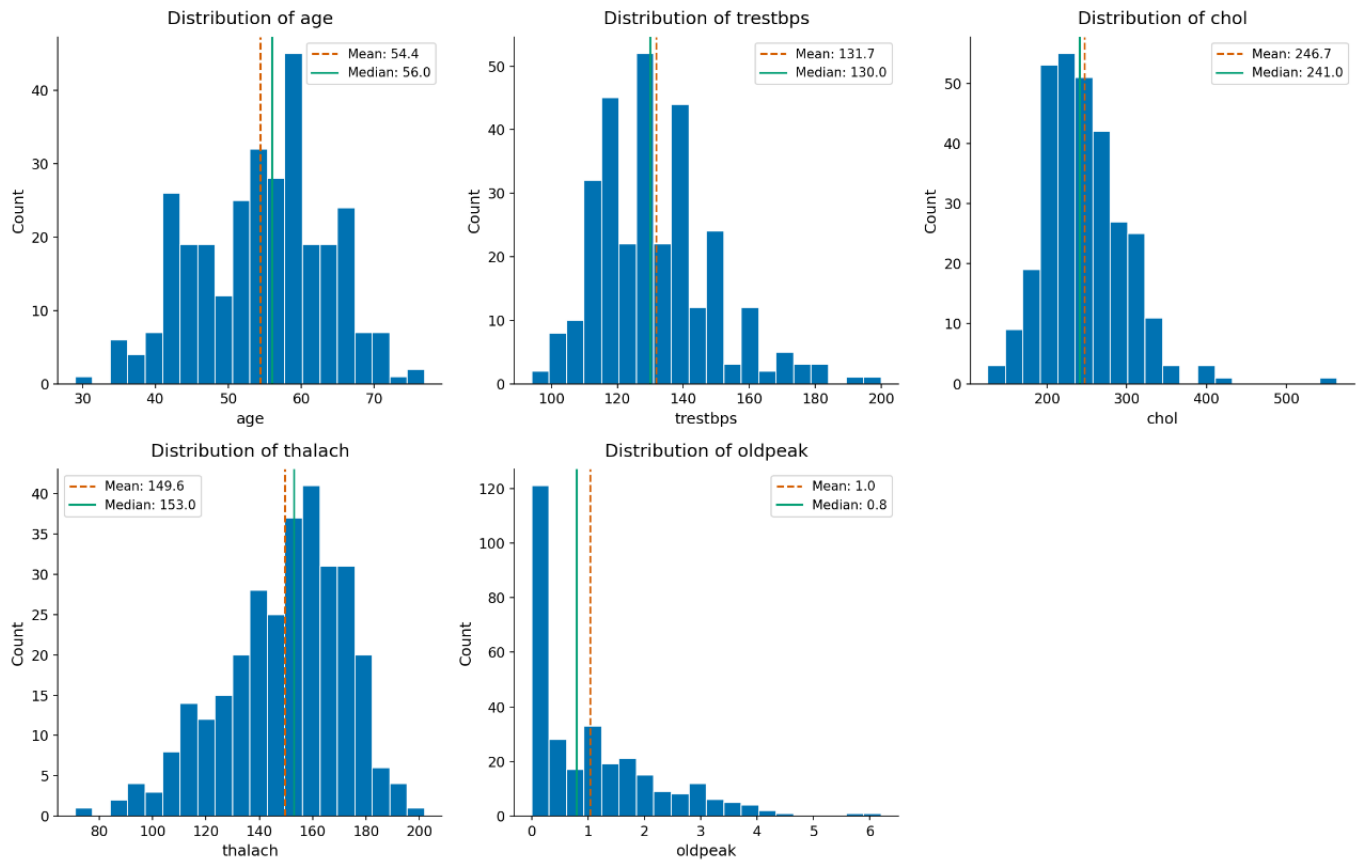


Figure 1: Distributional plots for all five continuous variables(age, trestbps, chol, thalach, oldpeak) . Orange dashed lines indicate mean values. Green solid lines indicate median values. Right skew observed in cholesterol and oldpeak, with mean pulled right by high value outliers. Age and thalach show more symmetric distributions.

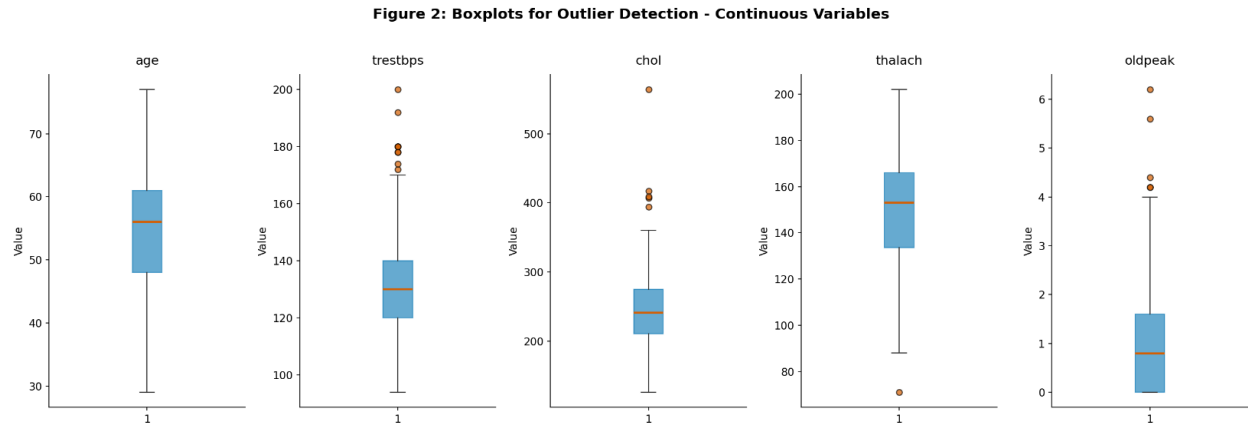


Figure 2: Boxplots for all five continuous variables. Orange horizontal line = median values. Orange dots = outliers. Cholesterol is showing an extreme outlier. Oldpeak shows multiple extreme outliers consistent with right skew identified in Figure 1. Resting blood pressure shows many outliers consistent with hypertensive patients. Age and maximum heart rate show minimal outliers.

	Variable	Category	Count	Proportion
0	sex	0.0	97	0.320
1	sex	1.0	206	0.680
2	cp	1.0	23	0.076
3	cp	2.0	50	0.165
4	cp	3.0	86	0.284
5	cp	4.0	144	0.475
6	fbs	0.0	258	0.851
7	fbs	1.0	45	0.149
8	restecg	0.0	151	0.498
9	restecg	1.0	4	0.013
10	restecg	2.0	148	0.488
11	exang	0.0	204	0.673
12	exang	1.0	99	0.327
13	slope	1.0	142	0.469
14	slope	2.0	140	0.462
15	slope	3.0	21	0.069
16	ca	0.0	176	0.581
17	ca	1.0	65	0.215
18	ca	2.0	38	0.125
19	ca	3.0	20	0.066
20	ca	NaN	4	0.013
21	thal	3.0	166	0.548
22	thal	6.0	18	0.059
23	thal	7.0	117	0.386
24	thal	NaN	2	0.007

Table 2. Frequency distributions for categorical variables including count and percentage by category.

**Figure 10: Logistic Regression Confusion Matrix
Test Set Results**

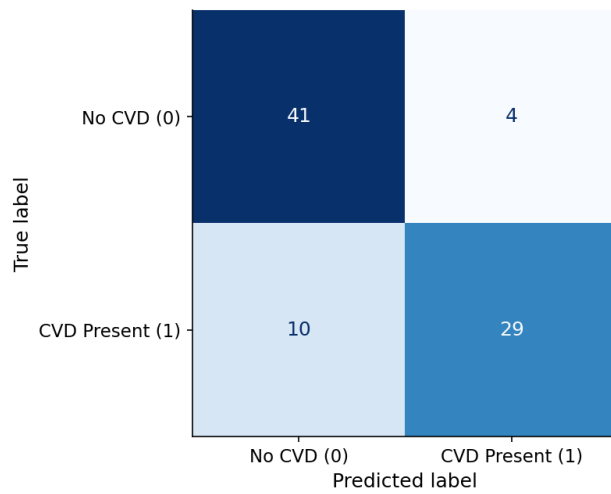


Figure 10: Confusion Matrix Logistic Regression Baseline Standard Model.

Figure 11: ROC Curve - Logistic Regression Baseline

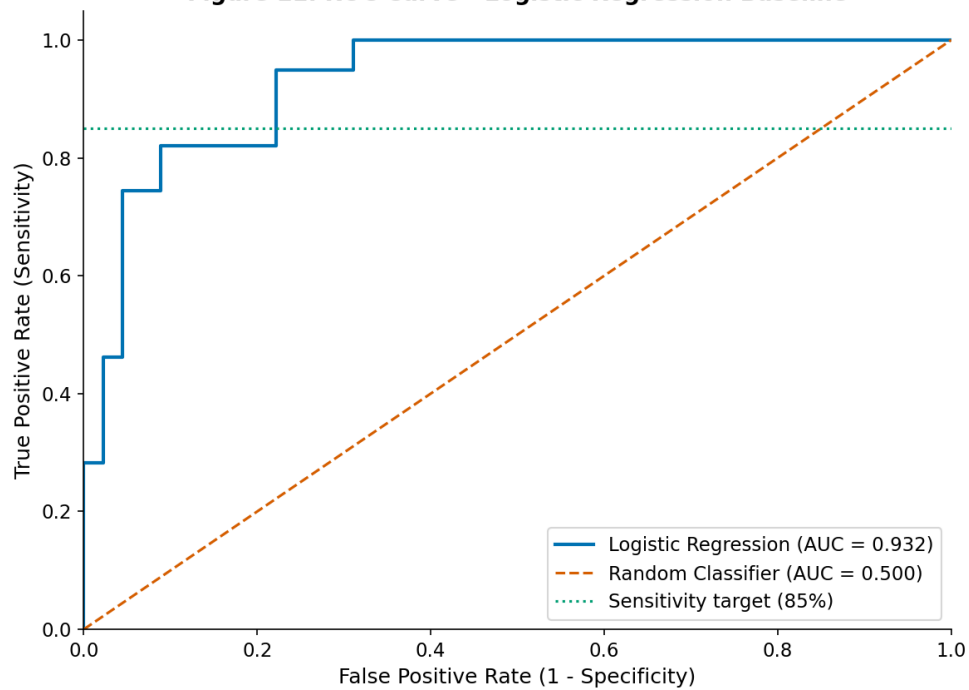


Figure 11: ROC Curve Logistic Regression Standard Baseline Blue solid line = Standard LR. Orange dotted diagonal = Random classifier. Green dotted = 85% sensitivity target

**Figure 13: Logistic Regression - SMOTE Confusion Matrix
Test Set Results**

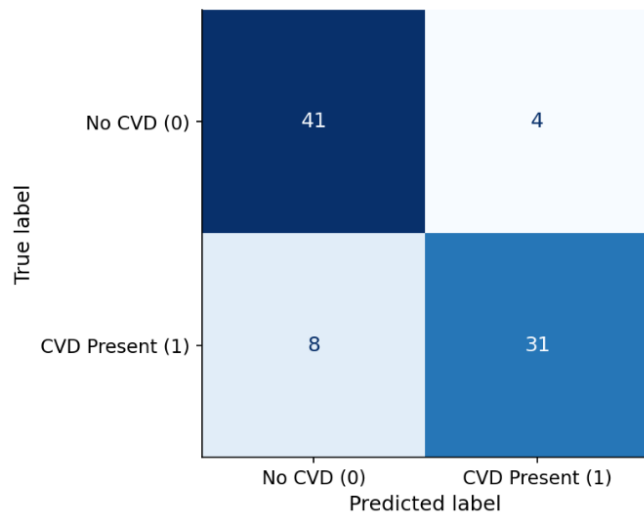


Figure 13: LR – SMOTE Confusion Matrix

**Figure 14: ROC Curve Comparison
Standard vs SMOTE Logistic Regression**

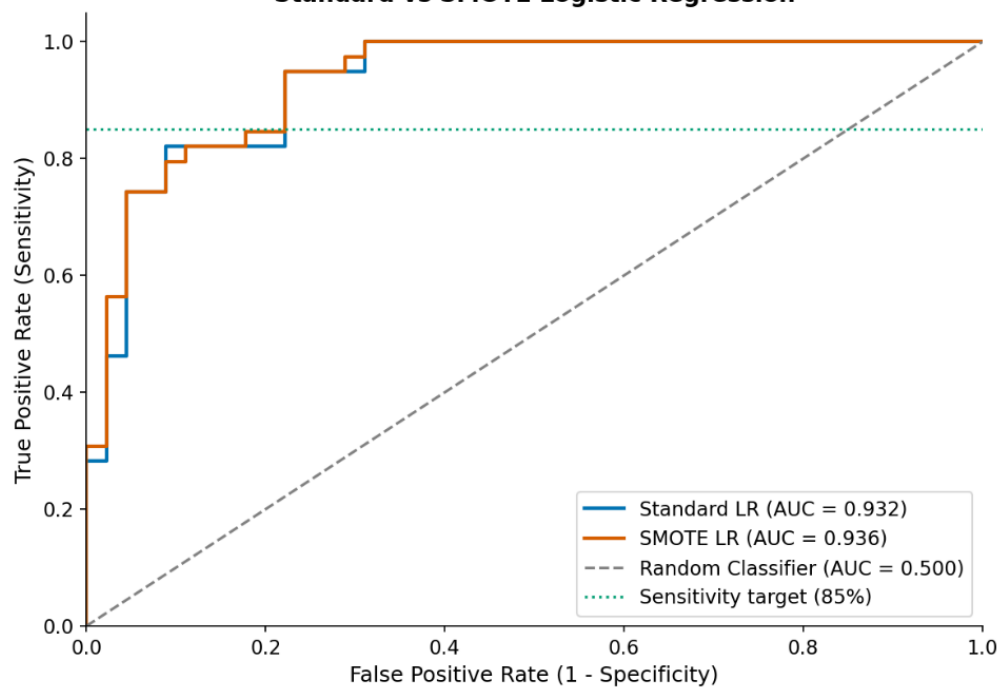


Figure 14: ROC Curve comparison of LR Standard vs SMOTE. Blue solid line = Standard LR. Orange solid line = SMOTE LR. Dotted diagonal = Random classifier. Green dotted = 85% sensitivity target

**Figure 15: Logistic Regression - CSL Confusion Matrix
Test Set Results**

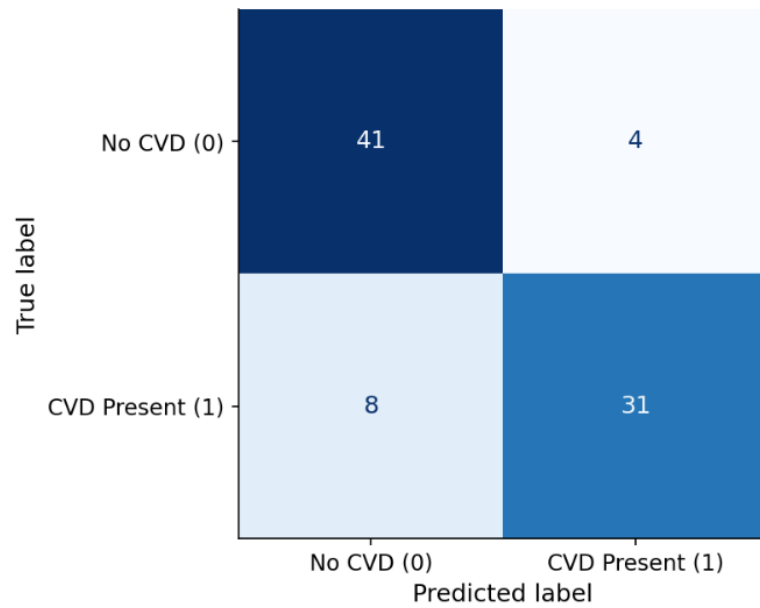


Figure 15: Confusion Matrix for LR CSL

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